WORKED SOLUTION

STEP III 1993, Question 8

Pure • Planes and Lines in 3D

STEP III 1993, Question 8

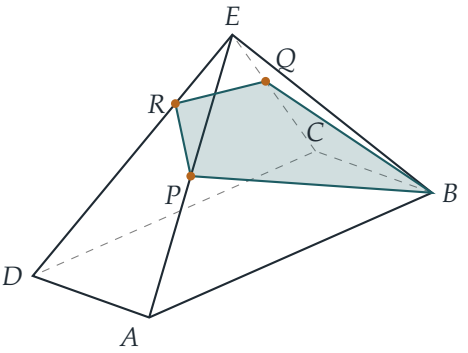
Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s3-q8

The key idea

Every point on a slant edge has a simple parametric form, because each edge joins a base vertex to E. So P, Q and R can all be written down at once, and the plane through B, P and Q can be found as a linear equation whose coefficients come from substituting those points. For the second part, the cleanest test for a parallelogram is that its diagonals bisect each other; one coordinate is enough to show this fails.

Solution



Finding γ

The edge AE consists of the points $(ta, 0, (1-t)a)$ for $0 \leq t \leq 1$, since $x + z = a$ along it. Similarly CE is $\{(-ta, 0, (1-t)a)\}$ and DE is $\{(0, -ta, (1-t)a)\}$. So

$$P = (\lambda a,\, 0,\, (1-\lambda)a), \qquad Q = (-\mu a,\, 0,\, (1-\mu)a), \qquad R = (0,\, -\gamma a,\, (1-\gamma)a).$$

We look for the plane BPQ in the form

$$px + qy + rz = a.$$

The points B, P, Q are not collinear, so they determine a unique plane. If we can find p, q, r satisfying all three conditions, that equation is the plane. Substituting each point:

$$\begin{aligned} B: &\quad qa = a, && \text{so } q = 1,\\ P: &\quad p\lambda + r(1-\lambda) = 1,\\ Q: &\quad -p\mu + r(1-\mu) = 1. \end{aligned}$$

To eliminate p, multiply the P equation by μ, the Q equation by λ, and add:

$$r(\mu - \lambda\mu + \lambda - \lambda\mu) = \lambda + \mu, \qquad \text{so} \qquad r = \frac{\lambda+\mu}{\lambda+\mu-2\lambda\mu}.$$

The denominator is $\lambda(1-\mu) + \mu(1-\lambda) > 0$, so this is fine. (We won't need it, but $p = (\lambda-\mu)/(\lambda+\mu-2\lambda\mu)$.)

Now R lies in the plane, so

$$-\gamma + r(1-\gamma) = 1 \quad\Longrightarrow\quad \gamma(1+r) = r-1 \quad\Longrightarrow\quad \gamma = \frac{r-1}{r+1}.$$

Putting everything over the common denominator $\lambda+\mu-2\lambda\mu$:

$$r-1 = \frac{2\lambda\mu}{\lambda+\mu-2\lambda\mu}, \qquad r+1 = \frac{2(\lambda+\mu-\lambda\mu)}{\lambda+\mu-2\lambda\mu},$$

and hence

$$\gamma = \frac{\lambda\mu}{\lambda+\mu-\lambda\mu},$$

as required.

It is worth noting that $0 < \gamma < 1$, so R really does lie on the edge DE strictly between D and E. Positivity is clear. For $\gamma < 1$, the denominator exceeds the numerator by $\lambda+\mu-2\lambda\mu = \lambda(1-\mu) + \mu(1-\lambda) > 0$.

$BPRQ$ is not a parallelogram

Ask yourself What is the quickest test for whether four points, taken in a given order, form a parallelogram?

A quadrilateral $BPRQ$ is a parallelogram exactly when its diagonals BR and PQ bisect each other, that is, when

$$\overrightarrow{OB} + \overrightarrow{OR} = \overrightarrow{OP} + \overrightarrow{OQ}.$$

(This is the same as $\overrightarrow{BP} = \overrightarrow{QR}$, i.e. one pair of opposite sides is equal and parallel.) Compare the y-coordinates of the two sides:

$$\text{left: } a - \gamma a = (1-\gamma)a, \qquad \text{right: } 0 + 0 = 0.$$

These would have to be equal, forcing $\gamma = 1$. But we showed $\gamma < 1$; indeed $\gamma = 1$ would put R at E. So the diagonals never bisect each other, and $BPRQ$ cannot be a parallelogram. ■

For completeness: the vertices really do occur in the order B, P, R, Q round the quadrilateral, because the plane cuts the four slant edges in the cyclic order EB, EA, ED, EC. Even pairing the points differently does not help. $B + P = Q + R$ and $B + Q = P + R$ would both need $a = -\gamma a$ in the y-coordinate, which is impossible since $\gamma > 0$.

A geometric way to see the same thing: BP lies in face ABE and QR lies in face CDE. These two faces meet in the line through E parallel to AB. Two parallel lines lying in these two planes would have to be parallel to that common line. So $BP \parallel QR$ would force $BP \parallel AB$, which is impossible since $P \neq A$.

Checks

- Symmetric case $\lambda = \mu$: the formula gives $\gamma = \lambda/(2 - \lambda)$. With $\lambda = \mu = \frac{1}{2}$ this is $\frac{1}{3}$. Directly: $p = 0$, $r = 2$ and the plane is $y + 2z = a$, which meets DE at $(0, -\frac{1}{3}a, \frac{2}{3}a)$, as predicted.
- Limit $\lambda, \mu \to 1$: $P \to A$, $Q \to C$, the plane becomes the base, and $\gamma \to 1$, so $R \to D$, as it should.
- A neat rewriting is $\dfrac{1-\gamma}{\gamma} = \dfrac{1-\lambda}{\lambda} + \dfrac{1-\mu}{\mu}$, which is easy to test on numbers. For example, $\lambda = \frac{1}{2}$, $\mu = \frac{2}{5}$ gives $\gamma = \frac{2}{7} \approx 0.2857$.
- We confirmed the formula, $0 < \gamma < 1$ and the parallelogram claim exactly (with fractions) on 2000 random choices of a, λ, μ. We found R independently, via a cross-product normal to the plane.

Common slips

- Parametrising AE wrongly: on AE the coordinates satisfy $x + z = a$ and $y = 0$, so P's z-coordinate is $(1 - \lambda)a$, not λa.
- Testing only that $BP \parallel QR$ by slopes in a 2D picture. The quadrilateral is in a tilted plane, so work with 3D vectors.
- Forgetting to show $\gamma < 1$, which is exactly what the parallelogram argument needs.

Beyond STEP

Finding where a plane cuts a line is the simplest case of solving a linear system. Since $EP : PA = (1 - \lambda) : \lambda$, and similarly on the other edges, the result says

$$\frac{ER}{RD} = \frac{EP}{PA} + \frac{EQ}{QC},$$

a coordinate-free statement about any plane through B. Because it uses only ratios along lines, it holds for every pyramid on a parallelogram base: an affine map takes any such pyramid to this one. Strang's *Introduction to Linear Algebra* develops planes, normals and linear systems properly.

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